

Danilov Sergey Dmitrievich

Saint Petersburg, Russia · +7 (986) 767-86-96 @geniusserg

danilovsergeyd@vk.com linkedin.com/in/geniusserg

> ML/LLM Inference Engineer — I own the full lifecycle of LLM inference optimization and reliability improvement. I research new SOTA algorithms, model and validate their impact, and deploy them to customer environments.

My tasks include:

- Choosing the best DP/TP strategy by modeling 10k-node inference clusters
- tuning replica scheduling and experimenting with different MoE EPLB placement strategies
- inference strategies considering the reliability of individual inference-cluster elements (memory, NPUs, network, software — vLLM, VeRL, Agents) based on statistical approaches and discrete-event simulations
- LLMOps methodologies for serving and deploying optimization code, including for use in multi-agent systems

CAREER

Huawei — LLM Inference R&D Engineer · *September 2024 – present*

- Developed topology-aware combinatorial EPLB MoE expert-placement algorithms that co-optimize expert-to-NPU mapping with all-to-all communication and fused kernels to minimize network congestion and tail latency — up to 10% lower network imbalance and 5% TPOT improvement (95% CI ±1%).
- LLMOps for an LLM quantization project: owned the full lifecycle from research notebook → quantized model → deployed service and benchmarks, speeding up the research cycle by 3× (PyTorch, vLLM, DeepEval).
- Disaggregated Prefill/Decode cluster simulator — Python discrete-event simulation of a distributed prefill/decode system with KV-cache-locality-aware load balancing and a Markov request-traffic generator; tested hypotheses for TP/DP/SP/EP topology, scheduling policy, and fast fault recovery to hold the customer's 99.99% SLO. The same simulator was used to test the RL (VeRL) rollout-worker recovery strategy, cutting recovery time from 10 to 5 minutes and keeping the TPOT drop under failures to no more than 5%.
- Designed a NUMA-aware memory stress test that reached 95% of theoretical bandwidth; modeled MTTR with a Weibull distribution and built an XGBoost failure-forecasting model that cut memory-related repair costs by 2×. doc
- Trained a time-series clustering model on optical-power telemetry that flags degrading links early; a minimal-parameter model (1M params) running on small chips raised cluster availability by +10%

with no signal-to-noise requirement — only raw optical-sensor data without link-failure labels. Used sktime for statistical and ML time-series analysis and PyTorch for model training.

- Integrated my projects as MCP tools into the internal Harness agent lab.

ML Engineer — ITMO University · *November 2023 – September 2024*

- Sber & ITMO: designed a CTGAN user emulator, used the synthetic data to train BiVAE for the VK feed, and improved MAP@5 by 3% on open data.
- Almazov & ITMO: applied my master's thesis on Shapley-value weighting of small datasets to train an XGBoost model for predicting obesity-treatment success; improved accuracy from 65% to 71% (+7%) for 3-month sibutramine therapy response; 2 Q2 papers. J. Pers. Med. J. Clin. Med.
- Industrial AI assistant: improved PDF NER with prompt engineering/self-consistency; filtering removed 80% repetition.

Testing Engineer — Harman · *September 2021 – July 2022*

- Sped up testing for complex GPS automotive systems with Roboto Framework

Release Engineer — Intel · *July 2019 – August 2021*

- Supported Intel OneAPI HPC release infrastructure: Python/Bash installation automation and desktop installer development.

TECH STACK

- **AI infrastructure:** Harness, Agents workflow, RAG, CAG, vLLM, VeRL, PyTorch, Kubernetes
- **Data Analysis & Research:** Python, Pandas, Scikit-learn, AutoML, PatentScope, Elsevier Scopus

EDUCATION

ITMO University, 2024 — Master's degree in Data Analysis

– Thesis: Shapley-value weighted filtering for small-sample ML; +7% accuracy on medical data.

Higher School of Economics, 2022 — Bachelor's degree in Software Engineering

– Thesis: accessible assistant for visually impaired users using Yandex Cloud OCR.

PUBLICATIONS

- Google Scholar · h-index 2
- Danilov et al., J. Pers. Med. 2024, 14, 811 — sibutramine therapy prediction.
- Markina et al., J. Clin. Med. 2024, 13, 4151 — statistical methods for studying brown adipose tissue and its effect on obesity treatment.

COMPETITIONS

Name That LLM — Kaggle competition

– End-to-end LLM text attribution pipeline: ModernBERT with SDPA, contrastive fine-tuning, classification by cosine similarity of embeddings. Leaderboard score: 0.3. Notebook

ScienceRAG — Research assistant agent based on GigaChat

– ScienceRAG: improved ArXiv RAG retrieval with a GigaChat-based agent. YouTube

X5 AI Tech Hackathon — Hallucination detection algorithm

– X5 AI Tech: ruBERT hallucination detector — 88% accuracy, <0.45 s inference; top-10.

LANGUAGES

- Russian, English (B1)